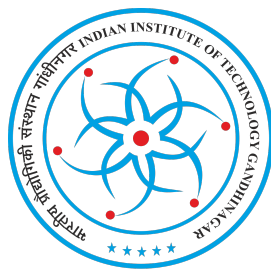


INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR (IIT Gandhinagar)



Assignment Report

M.Tech in Integrated Circuit Design and Technology (ICDT)

Name:	Kalaparthi Vikas (25210064)
Subject Name:	CMOS Analog IC Design
Subject Code:	EE 651
Assignment No:	05
Submission Date:	13 October 2025

1. Design an Operational **Transconductance Amplifier (OTA)** for the following specifications:

- Differential Gain = 40 - 60 dB (Gain should vary between 40 to 60 dB by changing V_{cont})
- CMRR > 70 dB
- ICMR: 0.80 V – 1.4 V
- Output Swing > 800 mV
- Power < 20 μ W
- Load Capacitance = 1 pF

Perform frequency analysis and calculate bandwidth of the designed amplifier. Plot the frequency response (both Gain and Phase) with the capacitive load.

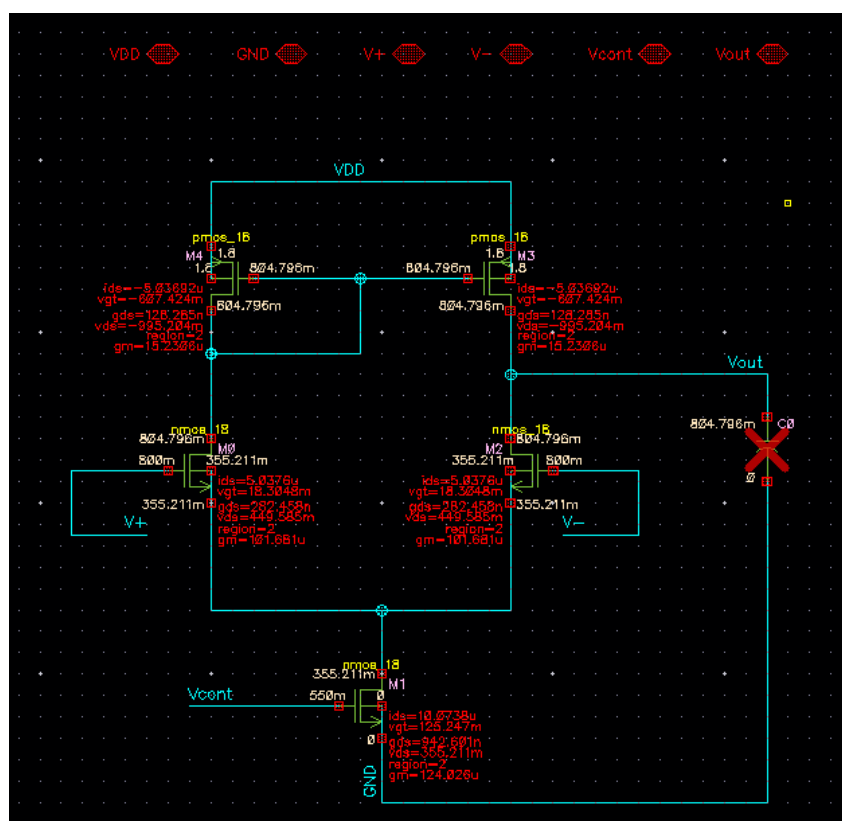


Figure 1: Schematic with DC parameters

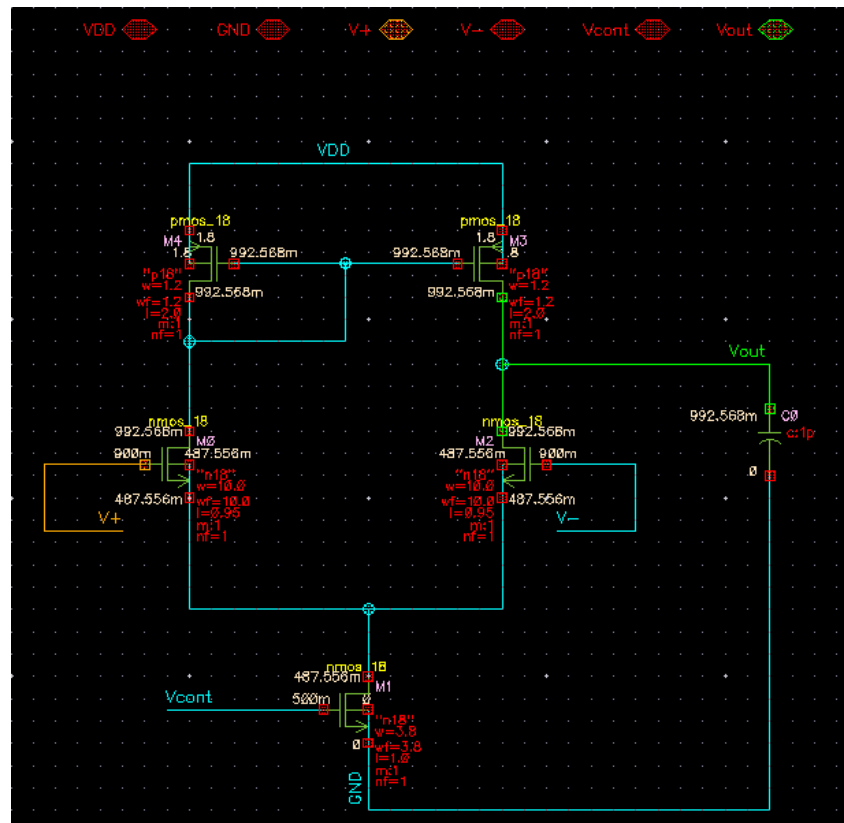


Figure 2: Schematic with Component Parameters

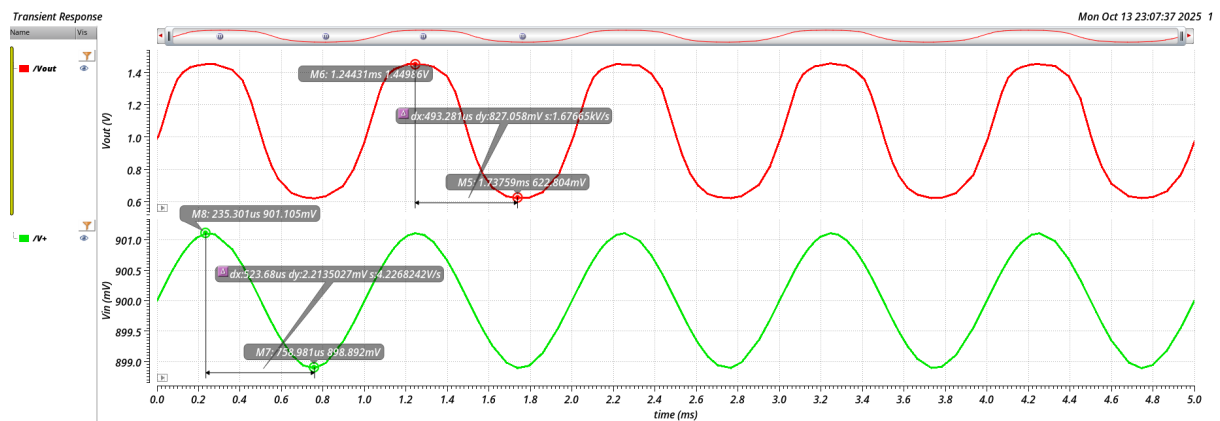


Figure 3: Trasient Analysis

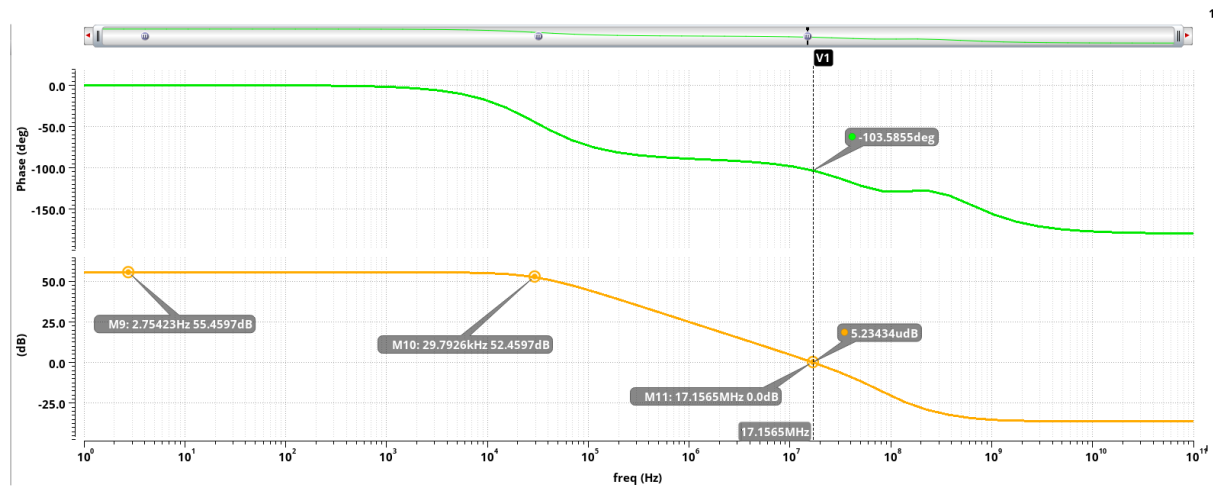


Figure 4: Gain and Phase Plot

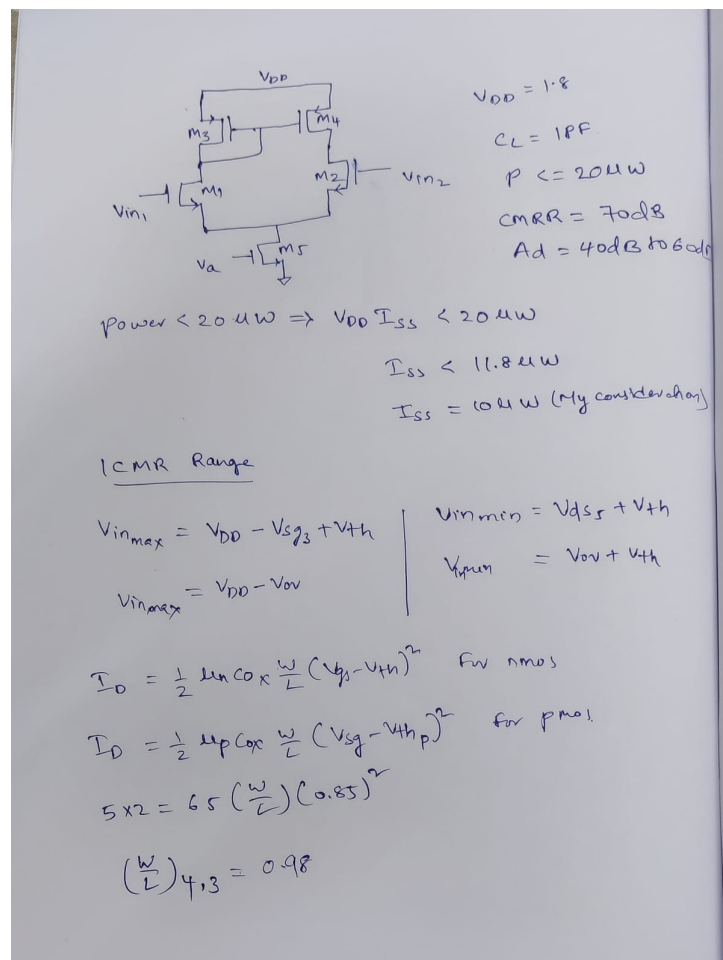


Figure 5: Theoretical Analysis

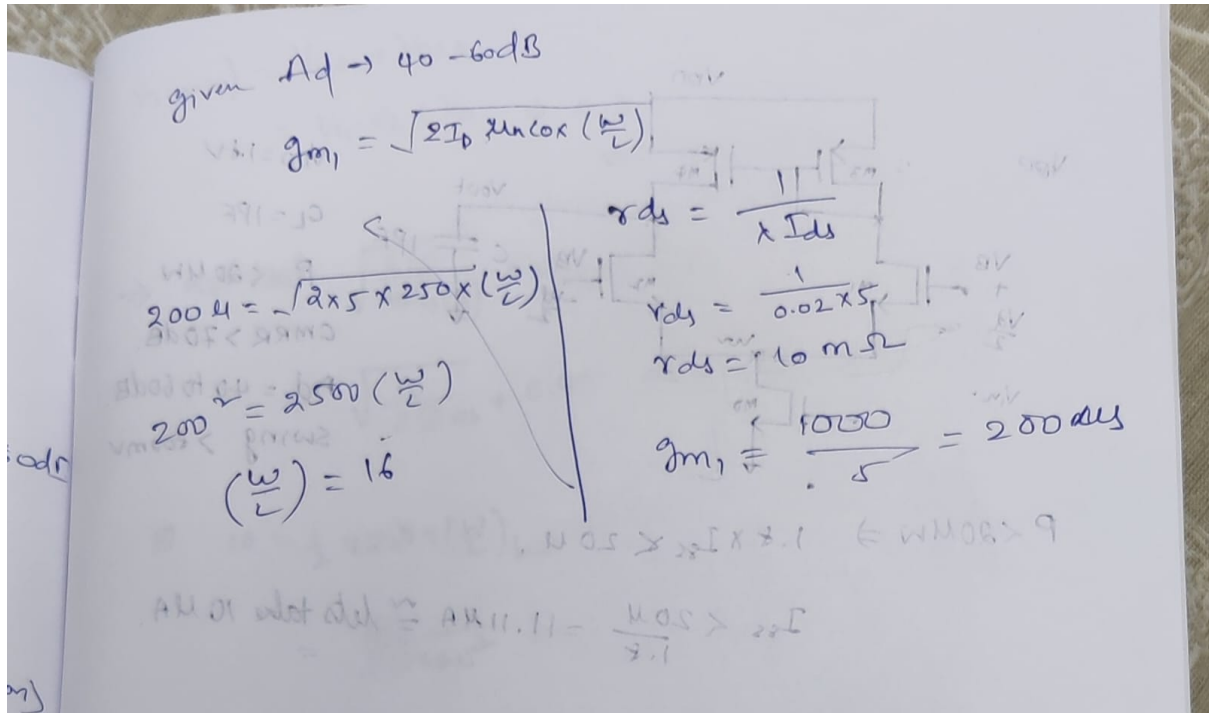


Figure 6: Theoretical Analysis

Observations

	Width (μm)	Length (μm)	Region	g_m (μS)	g_{ds} (μS)	V_{ov} (mV)
M3 = M4	1.2	2	2	15.2306	128.285	607.424
M1	5	3.8	2	124.026	942.601	125.247
M0 = M2	10	0.95	2	101.681	282.458	18.3064
V_{DD}	1.8	$V_{out}(\text{tail})$			I_{SS}	10.0738 μA
A_{DM}	48.1dB	A_{CM}		CMRR		Output Swing > 804mV
Phase Margin	77 degree					